

Chapter 1

Introduction

Symbolical artificial intelligence is a field of computer science that is highly related to quantum computation. At first glance, this statement appears to be a contradiction. However, the artificial intelligence framework, such as search and production system theory, allows an elegant description of a quantum computer model that is capable of quickly executing programs.

1.1 Artificial Intelligence

Artificial intelligence (AI) is a subfield of computer science that models the mechanisms of intelligent human behavior (intelligence). This approach is accomplished via simulation with the help of artificial artifacts, typically with computer programs on a machine that performs calculations. It should be noted that the machine does not need to be electronic. Indeed, Charles Babbage (1791-1871) sketched the first mechanical machine (a difference engine) for the calculation of certain values of polynomial functions [Hyman (1985)]. With the goal of mechanizing calculation steps, Babbage sketched the first model of a mechanical universal computer and called it an analytical engine. At the same time, Lady Ada Lovelance (1815-1852) thought about the computing power of such a machine. She argued that such a machine could only perform what it was told to do; such a machine could not generate new knowledge.

The term “artificial intelligence” itself was invented by the American computer scientist John McCarthy. It was used in the title of a conference that took place in the year 1956 at Dartmouth College in the USA. During this meeting, programs were presented that played chess and checkers, proved theorems and interpreted texts. The programs were thought to simulate human intelligent behavior. However, the terms “intelligence”

and “intelligent human behavior” are not very well defined and understood. The definition of artificial intelligence leads to the paradox of a discipline whose principal purpose is its own definition.

A.M. Turing (1912-1954), in 1950, wrote the essay “Computing Machinery and Intelligence”, in which he poses the question of how to determine whether a program is intelligent or not [Turing (1950)]. He defines intelligence as the reaction of an intelligent being to certain questions. This behavior can be tested by the so-called Turing test. A subject communicates over a computer terminal with two non-visible partners, a program and a human. If the subject cannot differentiate between the human and the program, the program is called intelligent. The questions posed can originate from any domain. However, if the domain is restricted, then the test is called a restricted Turing test. A restricted domain could be, for example, a medical diagnosis or the game of chess.

Human problem-solving algorithms are studied in Artificial Intelligence (AI). The key idea behind these algorithms is the symbolic representation of the domain in which the problems are solved. Symbols are used to denote or refer to something other than themselves, namely other things in the world (according to the, pioneering work of Tarski [Tarski (1944, 1956, 1995)]). They are defined by their occurrence in a structure and by a formal language which manipulates these structures [Simon (1991); Newell (1990)] (see Figure 1.1). In this context, symbols do not, by themselves, represent any utilizable knowledge. For example, they cannot be used for a definition of similarity criteria between themselves. The use of symbols in algorithms which imitate human intelligent behavior led to the famous physical symbol system hypothesis by Newell and Simon (1976) [Newell and Simon (1976)]: “The necessary and sufficient condition for a physical system to exhibit intelligence is that it be a physical symbol system.” Symbols are not present in the world; they are the constructs of a human mind and simplify the process of representation used in communication and problem solving.

1.1.1 *Machine Learning*

There are some parallels between humans (animals) and machine learning. During learning, humans attempt to gain some knowledge, which involves some modification of behavioral tendencies by experience. Many of the techniques are derived from the efforts of psychologists and biologists to make more sense of human learning through computational models

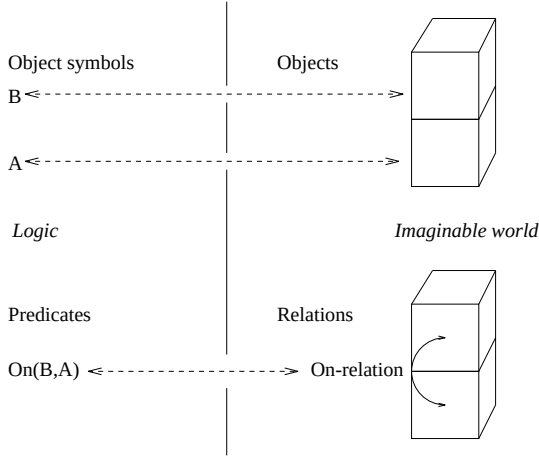


Fig. 1.1 Object represented by symbols and relation represented by predicate.

[Anderson (1995b)]. In machine learning a sub-symbolical or distributed representation is used. One form of distributed representation corresponds to a pattern that mirrors the way the biological sense organs describe the world. Perception organs sense the world by receptors. By the given order of the receptors, living organisms experience reality as a vector in which each dimension corresponds to a certain value in the receptor [Wichert (2009)]. One can measure the distance between two D dimensional vectors by a distance function like the the Euclidean distance function

$$d(\mathbf{x}, \mathbf{y}) = \|\mathbf{x} - \mathbf{y}\| = \sqrt{|x_1 - y_1|^2 + |x_2 - y_2|^2 + \dots + |x_D - y_D|^2}.$$

We can describe objects by features. Let us imagine that we want to describe two species of a fish, the sea bass and the salmon, by some features (see Figure 1.2), [Duda *et al.* (2000)]. The process of choosing the correct features is called feature extraction. In our example, only two features are chosen since they can be inspected with the aid of a two-dimensional coordinate system. The two chosen features are width and lightness. In Figure 1.3, each salmon is represented by a dot and each sea bass by a cross. We attempt to separate the two classes by a line; we have some errors, but most examples are correctly separated. One of the most important machine learning algorithms, perceptron, attempts to determine (learn) this line from a given sample. Later, during classification, an unknown fish is described by a two-dimensional vector, and depending on which side of the separating line it falls, it is classified as either salmon or sea bass.

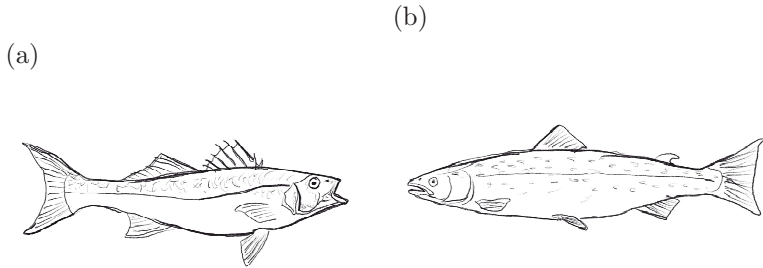


Fig. 1.2 (a) Sea bass and (b) salmon.

In machine learning, we can distinguish between supervised learning, as described in the preceding example and unsupervised learning. In supervised learning, the algorithm is presented with examples of inputs and their desired outputs. The goal is to learn a general rule that maps inputs to outputs. Supervised learning is frequently referred to as learning with a teacher because the desired outputs are indicated by a type of teacher. Consequently, unsupervised learning is referred to as learning without a teacher. In unsupervised learning, the algorithm groups information that is primarily represented by vectors into groups. The algorithm attempts to find the hidden structure of unlabeled data; clustering is an example of such an algorithm.

Deep artificial neural networks have many hidden layers usually trained by backpropagation. The models achieved tremendous results, and it is now common to identify artificial intelligence with deep learning and not with symbol manipulating systems. This results from the paradox of artificial intelligence as stated before in which the terms “intelligence” and “intelligent human behavior” are not very well defined and understood.

1.2 Motivation and Goals

Traditional AI is built around abstract algorithms and data structures that manipulate symbols. One of the important algorithms is the tree or graph search. Common forms of knowledge representation are symbolic rules and semantic nets. Traditional AI attempts to imitate human behavior without any relationship to physical reality, for example, in biological hardware.

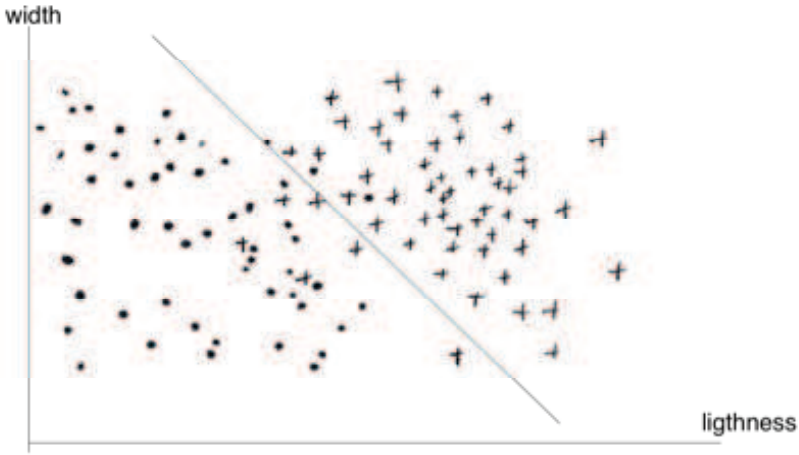


Fig. 1.3 In a sample, each salmon is represented by a dot and each sea bass by a cross. We separate the two classes by a line. Some errors are present, but most examples are correctly separated. Later, during classification, an unknown fish is described by a two-dimensional vector, and depending on which side of the separating line it falls, it is classified as either salmon or sea bass.

Sub-symbolical processing, on the other hand, belongs to biology-inspired AI, which involves methods such as behavioral systems or neural networks and statistical machine learning. Could the physical nature, as described by quantum physics, also lead to algorithms that imitate human behavior? What are the possibilities for the realization of artificial intelligence by means of quantum computation? Computational algorithms that are inspired by this physical reality are described by quantum computation. We will answer questions such as why and how to use quantum algorithms in artificial intelligence.

Questions that appear to be quite simple, such as: what are random numbers and how can we generate them, cannot be answered by traditional computer science. The widely used pseudo random generators are based on deterministic procedures and do not generate randomness; instead, they generate pseudo-randomness. Pseudo random generators are related to deterministic chaos sequences, which are described by mathematical chaos theory. Chaotic patterns can arise from very simple mathematics. While

the results can be similar to true randomness, the output patterns are generated by deterministic rules. Chaotic patterns differ from most deterministic systems because any small change made to their variables can result in unpredictable changes to the system behavior.

In this book, we introduce quantum computation and its application to AI. Based on information science, we will illustrate the general principles that govern information processing and information structures. Artificial intelligence (AI) can be divided into different areas: symbolical artificial intelligence, machine learning, cognitive psychology and philosophy. We demonstrate how quantum computation and quantum physics contribute to each of these subareas.

Quantum algorithms for AI have been proposed by [Tarrataca and Wichert (2011b,a, 2012b, 2013b)], including a quantum tree search algorithm and a quantum production system.

Linear algebra-based quantum machine learning is based on the HHL quantum algorithm for solving linear equations, quantum PCA, and quantum kernels. Quantum annealing is demonstrated on a Boltzmann machine and deep belief networks. Quantum cognition leads to quantum Bayesian networks and quantum-like evolution models. Finally, we describe probabilities and the mind in the Everett many-worlds theory.

1.3 Guide to the Reader

This book is about some core ideas of artificial intelligence and quantum computation and is composed of four sections: the first is on classical computation and the second section is on quantum computation. In the first section, we introduce the basic principles of computation, representation and problem solving. Quantum physics is based on information theory and probability theory. We present both theories and indicate their relationships to artificial intelligence by associative memory and Bayesian networks. In the second section, we introduce the principles of quantum computation and its mathematical framework. We present two principles on which quantum computation is based, the discrete Fourier transform and Grover's algorithm. Based on these principles, we introduce the iterative quantum tree search algorithm that speeds up the search. In the next step we introduce a quantum production system on which a universal quantum computer model is based. In the third section, quantum machine learning is explained from

two perspectives, linear algebra-based quantum machine learning (e.g., the HHL quantum algorithm for solving linear equations) and quantum annealing for a Boltzmann machine and deep belief networks. The fourth section deals with quantum cognition and quantum-like models. Finally, the fifth section discusses the philosophical question of quantum computation and the multiverse. Readers who want to develop a general understanding of the quantum computation mathematical framework should read the second section, beginning with the chapter “Introduction to Quantum Physics?”. Sections that go into more detail are marked by a star “*” and can be skipped on the first reading.

1.4 Content

1.4.1 *Classical computation*

Computation - Chapter 2 The Entscheidungsproblem is presented, and the Turing machine is introduced. The proof of the Entscheidungsproblem is based on Cantor’s diagonal argument and Gödelization. The Universal Turing machine is an abstract model of a computer. Computational complexity theory addresses questions regarding which problems can be solved in a finite amount of time on a computer. The Church-Turing thesis states that any algorithmic process can be simulated on a Turing machine. Two classes of practical computers are presented: analog and digital computers.

Problem Solving - Chapter 3 In the first step, the knowledge representation framework is introduced: rules, logic-based operators, frames and categorical representations. In the next step, production systems are introduced. A production system is a model of human problem solving. It is composed of long-term memory and working memory, which is also called short-term memory. We distinguish between deduction systems and reaction systems. Planning can be performed more easily by reaction systems in which the premise specifies the conditions that must be satisfied; in this way, the condition that specifies an action can be undertaken. An 8-puzzle example is presented. There is an assumption that the distance between states in the problem space is related to the distance between the sub-symbols that represent the states in sub-symbolical problem-solving. Sub-symbolical problem-solving takes advantage of the geometric nature of the world.

Information - Chapter 4 Information theory is highly related to mathematical probability theory and thermodynamics. Entropy is a measure of the disorder of the configuration of states and can be described by a dice model. The Maxwell paradox identifies information with a negative measure of entropy. The ideal entropy represents the minimal number of optimal questions that must be addressed to know the result of an experiment. We will indicate the relationships between information and hierarchical structures and measurement. In the section on information and memory, we will introduce a biologically inspired model of associative memory. The Information and storage capacity of the model is high, given that the binary representation is sparse. Finally, a deduction system based on associative memory is presented.

Reversible Algorithms - Chapter 5 Bennett (1973) showed that irreversible overwriting of one bit causes at least $k \cdot T \cdot \log 2$ joules of energy dissipation, where k is Boltzmann's constant and T is the absolute temperature. Bennett also indicated that this lower bound can be ignored when using reversible computation. Reversible computing is a model of computing in which the computational process is reversible. Reversible Boolean gates and circuits are described.

Probability - Chapter 6 Probability theory is built around Kolmogorov's axioms. For a joint distribution of n possible variables, the exponential growth of combinations being true or false becomes an intractable problem for large n . There are two possible solutions to this problem, Naïve Bayes and Bayesian networks. Naïve Bayes is related to counting and categorization. For the unobservable variables, the Bayesian networks are based on the law of total probability. A Markov chain is a mathematical system that undergoes transitions described by a stochastic matrix. A stochastic matrix evolution that occurs when describing the evolution of a physical system is usually non-reversible. A vector representation of events for the possible worlds approach is introduced. In the vector based framework for probabilities a state represents events in a n -dimensional vector space by a unit-length vector in the norm l_1 . By replacing the l_1 norm by the Euclidean l_2 norm we introduce quantum probabilities.

1.4.2 Quantum computation

Introduction to Quantum Physics - Chapter 7 The unitary deterministic evolution represented by the Schrödinger equation and two inter-

pretations of quantum mechanics are presented. We indicate the difference between stochastic Markov evolution and unitary evolution. The mathematical framework of quantum theory is based on linear algebra in Hilbert space. The relationships between the unitary operators represented by unitary matrices and the Schrödinger equation and the Hamiltonian are described by a spectral representation. A 2-state quantum system is described by a two-dimensional Hilbert space. Such a 2-state quantum system corresponds to a qubit. A register is composed of several qubits and is defined by the tensor product. Shannon entropy of a superposition of qubits measures the distribution of the probabilities. It describes the departure of the state from a basis state (eigenstate). For a basis state, there is no uncertainty during the measurement. The higher the entropy is, the higher the uncertainty during the measurement.

Computation with Qubits - Chapter 8 A unitary operator on a qubit is called a unary quantum gate. An operation on several qubits can be represented by unitary matrices. The matrix representation of serial and parallel operations is composed of either a product or a tensor operation between the matrices. A reversible circuit of m bits corresponds to a unitary mapping. A reversible circuit can be represented by a unitary permutation matrix or by quantum Boolean gates. The Deutsch algorithm exploits the superposition of qubits that are generated by Hadamard gates; the Deutsch algorithm is more powerful than any other classical algorithm and determines whether an unknown function of one bit is constant or not by calling the function one time. A classical algorithm requires two calls. The Deutsch Jozsa algorithm generalizes the working principle even to a more powerful algorithm for a function of m bits.

Periodicity - Chapter 9 The Fourier transform changes a signal from the time domain to the frequency domain. The discrete Fourier transform changes discrete time-based or space-based data into frequency-based data. The discrete Fourier transform (DFT) can be seen as a linear transform represented by a unitary matrix F . This unitary matrix F also defines the quantum Fourier transform (QFT). The decomposition of the matrix is described by the fast Fourier transform (FFT). The QFT decomposition is equivalent to the FFT. The QFT period algorithm determines the period of a periodic function in polynomial time and is the basis of Shor's Algorithm for the factorization of numbers in polynomial time. An alternative approach is described in Kitaev's phase estimation algorithm, which

determines the eigenvalue for a unitary operator and an eigenvector.

Search - Chapter 10 We want to find x for which $f(x) = 1$, $x = \xi$. This task is equivalent to a decision problem with a binary answer $1 = \text{yes}$ and $0 = \text{no}$ and the instance x . There is a lower bound for a quantum search on a quantum computer using a quantum oracle. The possible speedup is quadratic; $NP - \text{complete}$ problems remain $NP - \text{complete}$. The search is described by Grover's algorithm and is based on the Householder reflection. In the case, the number of the solutions is unknown, we can apply a quantum counting algorithm. The generate-and-test method is a simple AI paradigm that can directly benefit from Grover's algorithm.

Quantum Problem-Solving - Chapter 11 Problem-solving can be modeled by a production system that implements a search algorithm. The search defines a problem space and can be represented as a tree. In an uninformed search, no additional information about the states is given. A heuristic search is based on a heuristic function $h(\nu)$ that estimates the cheapest cost from the node ν to the goal. However, inventing heuristic functions is difficult. An alternative approach is that of the quantum tree search algorithm. Using Grover's algorithm, we search through all possible paths and verify, for each path, whether it leads to the goal state. We present the iterative quantum tree search, which is the basis of the quantum production system. We explain the principles of Tarrataca's quantum production system on a trivial example, the 3-puzzle. Finally, we present a universal quantum computer model that is capable of more quickly executing programs. The corresponding principles can be integrated into the unified theories of human cognition.

1.4.3 Quantum machine learning

Grover's Algorithm and the Input Problem - Chapter 12 The input and the destruction problem are introduced. The amplitude distribution of a quantum state is initialized by reading N data points. This cost can dominate the complexity of an algorithm. A quantum memory can be queried only once because of its collapse during the measurement (destruction). We present a quantum algorithm based on Wilshaw's associative memory that efficiently prepares the data. The idea is to generate quantum superposition from the classical data where the reading cost is negligible.

Statistical Machine Learning - Chapter 13 In unsupervised learning, algorithm groups input information that is primarily represented by vectors into groups without additional information. The algorithm attempts to find the hidden structure of unlabeled data; principal component analysis is an example of such algorithm. In supervised learning, the algorithm is presented with example inputs and their desired outputs, and the goal is to obtain a general rule that maps inputs and outputs. We describe the principal component analysis, linear and logistic regression, backpropagation algorithm, support vector machines, and kernel machines.

Linear Algebra-based Quantum Machine Learning - Chapter 14

Linear algebra-based quantum machine learning is based on quantum gates that describe quantum basic linear algebra subroutines. These subroutines exhibit theoretical exponential speedups compared to classical counterparts and are essential for machine learning. The quantum algorithm for linear systems of equations is one of the main fundamental algorithms expected to provide a speedup compared to classical counterparts. The algorithm is also called the HHL algorithm and is based on Kitaev's phase algorithm. We describe the quantum principal component analysis (qPCA) and the quantum random access memory (qRAM). We introduce quantum kernels and indicate quantum advantage kernels.

Stochastic Methods - Chapter 15

We introduce the Hopfield and Ising models and describe the finite temperature dynamics. The stochastic dynamics of the Ising model can be described by Gibbs sampling. The metropolis algorithm is a modified Monte Carlo method; it is referred to as the Markov chain Monte Carlo. The rate of convergence of the Markov chain of the Gibbs sampling or the Metropolis algorithm to thermal equilibrium is extremely slow at very low temperatures. For improved computational efficiency, we operate the stochastic system at high temperature (where convergence to equilibrium is fast) and then maintain the system at equilibrium as the temperature is slowly lowered. Simulated annealing is named by analogy to real annealing that a real metallurgist does by slowly cooling the sample. The Boltzmann machine is a stochastic Hopfield network with hidden units. It is a stochastic binary machine whose composition consists of stochastic neurons in states 1 or 0. The learning process in the Boltzmann machine is very slow, particularly when the number of hidden units used in the machine is large. A restricted Boltzmann machine (RBM) has connections only between visible and hidden units to simplify

inference and learning. A deep belief network (DBN) is a generative graphical model or, alternatively, a class of deep neural networks composed of multiple layers of latent variables (hidden units) with connections between the layers but not between units within each layer. It can be viewed as a composition of restricted Boltzmann machines. Deep belief networks, which are also called deep neural networks (DNN) (Hinton, 2006), capture higher-level representations of input features.

Adiabatic Quantum Computation and Quantum Annealing - Chapter 16 Adiabatic quantum computation is an alternative approach to quantum computation. It is based on the time evolution of a quantum system. Adiabatic quantum computation is a polynomial equivalent to the quantum gate model. Adiabatic quantum computation and quantum annealing are frequently used interchangeably; however, there are some distinctions. Adiabatic quantum computation refers to a universal model that solves any Turing-compatible problems that run in an adiabatically closed system without interference from the environment. Quantum annealing solves optimization problems. In simulated annealing, the neighborhoods remain the same throughout the search. The temperature determines the probability of moving out of a local minimum. In the beginning, the temperature is high, and the probability of moving out of a minimum is high. Then, it is slowly reduced until the probability of moving out of a minimum is zero. The disordering Hamiltonian introduces quantum fluctuations in quantum annealing. The transverse field coefficient, which is also called the tunneling coefficient, controls the passage of the solution landscape by quantum tunneling. We introduce the D-Wave quantum annealer. An alternative approach for a Boltzmann machine or deep belief network is to estimate model expectations by a quantum annealer such as D-Wave. Variational algorithms are related to the adiabatic quantum computation; however, they are based on discrete unitary evolution. A variational quantum eigensolver is a hybrid algorithm based on a quantum device and a classical algorithm to estimate the ground state of a system.

1.4.4 *Quantum-like models*

Quantum Cognition - Chapter 17 Quantum cognition uses a mathematical quantum theory to model cognitive phenomena. It is assumed that the computation itself is performed on a classical computer and not on a quantum computer. In a quantum world, the brain is considered to be a

classical computer. The quantum probabilities, when observed, correspond to classical probability theory. If not observed, the state of a system is described by a complex vector with a length of one. Two equivalent states represent the same state when a measurement is performed, but they can behave differently during the unitary evolution. Humans, when making decisions, violate the law of total probability. The violation can be explained as a quantum interference. We introduce the law of balance and quantum-like Bayesian networks.

Quantum-like Evolution - Chapter 17 Evolutionary dynamics have traditionally been studied in the context of homogeneous populations. Population structures are modeled by arranging individuals on a graph in which each vertex represents an individual. Their evolutionary behavior is modeled by Markov chains described by a stochastic matrix. An important question is whether more generalizable probabilistic models can be found using quantum probabilities. Quantum-like evolution is l_2 norm preserving in which the matrix does not need to be unitary. By coding the phase of the input vector into the phase of the balanced phase matrix, we can simulate the unitary evolution. The phase that describes the matrix is dynamically updated. Quantum-like evolution can simulate either unitary evolution by coding the phase of the input vector into the phase of the balanced phase matrix, a Markov chain, or an alternative evolution that can lead to fixed, periodic, or chaotic behavior, which results in strange oscillations.

1.4.5 *Quantum philosophy*

Quantum Computation and the Multiverse - Chapter 18 The Everett many-worlds theory views reality as a many-branched tree in which every possible quantum outcome is realized. The modern version of the wave-function collapse is based on decoherence and leads to the multiverse interpretation of quantum mechanics. Following a controversial suggestion by David Deutsch that decision theory can solve the problem of probabilities in the Everett many-worlds, we suggest that the probabilities are induced by Shannon's entropy that measures the uncertainty of events. We argue that a relational person prefers certainty to uncertainty due to the fundamental biological principle of homeostasis. Then, we redefine free will as the concept of identity in a deterministic universe. Causality is related to a meaningful explanation. For impossible explanations, causality does not exist, and the identity of the self breaks. Personal identities can ex-

ist only in meaningful causal worlds. We are one person described by the meaningful path in the multiverse that is correlated with our free choices.

Conclusion- Chapter 20 In classical physics, randomness does not exist; in quantum physics, it cannot be explained. In addition to randomness, quantum computation cannot offer any noncomputable schema such as that claimed by the quantum mind or the quantum consciousness hypothesis. However, quantum cognition assumes that the brain is a classical computer in a quantum world. Because a wave function can be present at the macro scale, predictions, such as the hidden inference, are based on von Neumann probabilities and not on the classical probability theory.